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9702/32

May/June 2012

2 hours

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

You will be allowed to work with the apparatus for a maximum of one hour for each question.
You are expected to record all your observations as soon as these observations are made, and to plan the presentation of the records so that it is not necessary to make a fair copy of them.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
Total	

This document consists of **10** printed pages and **2** blank pages.

You may not need to use all of the materials provided.

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- 1** In this experiment, you will investigate the equilibrium position of a suspended mass.

The apparatus has been set up as shown in Fig. 1.1.

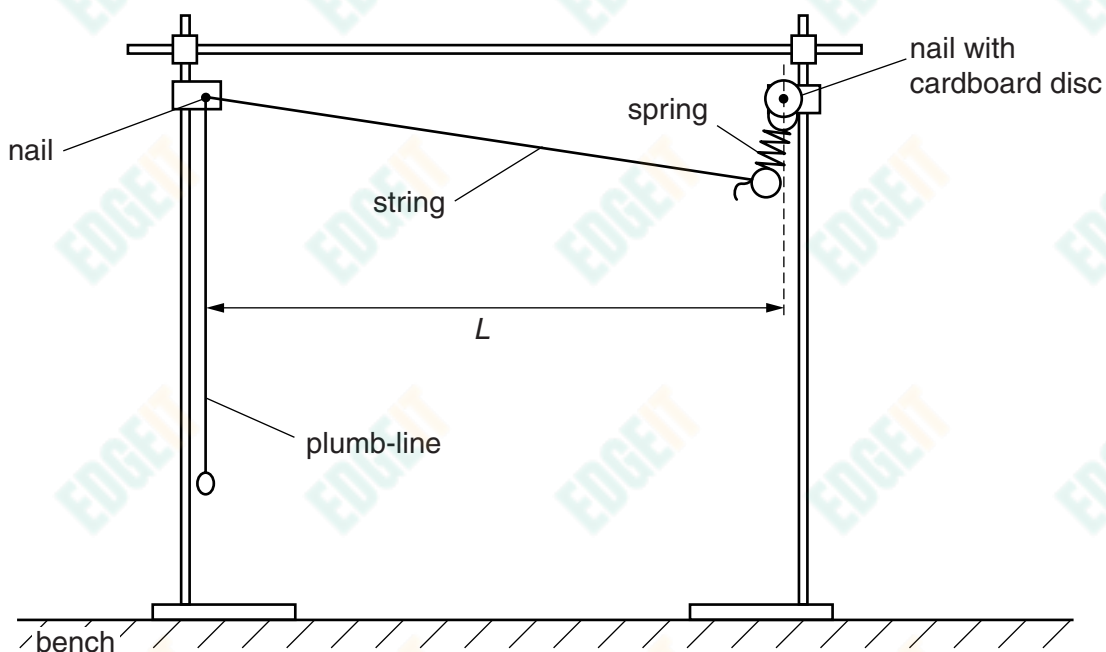


Fig. 1.1

- (a)** Measure and record the distance L between the two nails.

$L = \dots\dots\dots$ m [1]

- (b) (i)** Attach the crocodile clip to the string so that the string passes through the gap between the jaws of the crocodile clip, as shown in Fig. 1.2.

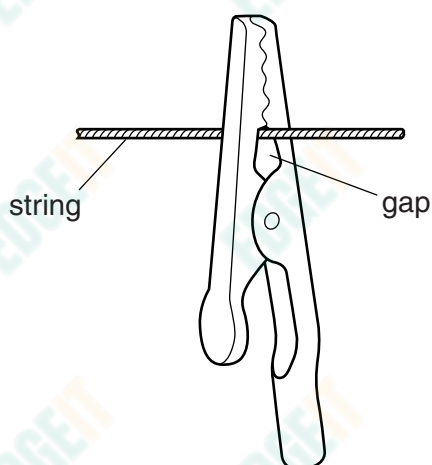


Fig. 1.2

Suspend the mass hanger and masses from the loop attached to the crocodile clip. Adjust the position of the crocodile clip so that the distance d from the plumb-line to the string loop is about 30 cm, as shown in Fig. 1.3.

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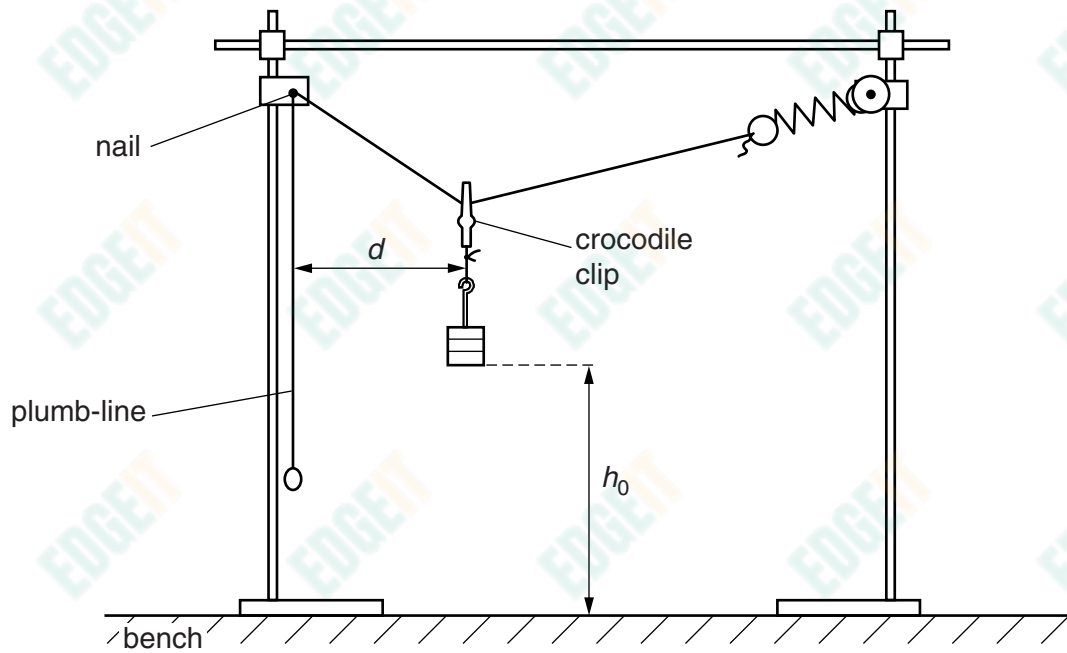


Fig. 1.3

- (ii) Measure and record d .

$d = \dots\dots\dots$ m

- (iii) Measure and record the initial height h_0 of the bottom of the mass above the bench, as shown in Fig. 1.3.

$h_0 = \dots\dots\dots$ m [1]



- (c) Reduce d by moving the crocodile clip **closer** to the plumb-line, ensuring that the string passes through the gap as shown in Fig. 1.2. Measure and record d and the new height h of the bottom of the mass above the bench. Repeat until you have six sets of values for d and h , with d in the range 5 cm to 30 cm.

Include values of $\left(\frac{L}{2} - d\right)^2$ and $(h - h_0)$ in your table.

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[10]

- (d) (i) Plot a graph of $\left(\frac{L}{2} - d\right)^2$ on the y -axis against $(h - h_0)$ on the x -axis.

[3]

- (ii) Draw the straight line of best fit.

[1]

- (iii) Determine the gradient and y -intercept of this line.

gradient =

y -intercept =

[2]

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- (e) The relationship between d and h is

$$\left(\frac{L}{2} - d\right)^2 = a(h - h_0) + b$$

where a and b are constants.

Using your answers from (d)(iii), determine the values of a and b .
Give appropriate units.

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$a =$

$b =$

[2]

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Please turn over for Question 2.

You may not need to use all of the materials provided.

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2 In this experiment, you will investigate the adhesive strength of Blu-Tack.

- (a) (i) You have been provided with two plastic rulers, one with a loop of string attached. Take the ruler without string and clamp it along the edge of the bench with a flat face uppermost.
- (ii) Use some of the Blu-Tack to make a ball of diameter about 8 mm.
- (iii) Using the calipers provided, measure and record the diameter of the ball.

ball diameter =

- (iv) Place the ball of Blu-Tack on the clamped ruler with a thin rod each side of it. Press the flat face of the second ruler on to the Blu-Tack until the ruler touches the rods. The Blu-Tack will form a disc, as shown in Fig. 2.1.

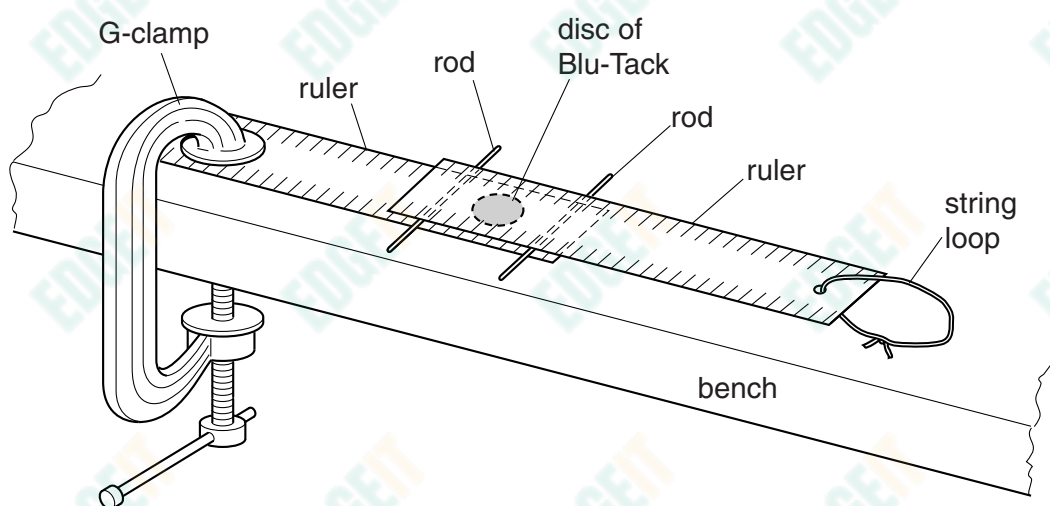


Fig. 2.1

- (b) (i) Measure and record the diameter d of the disc of Blu-Tack.

$d =$ [2]

- (ii) Estimate the percentage uncertainty in d .

percentage uncertainty = [1]

- (iii) Calculate the contact area A using $A = \frac{\pi d^2}{4}$.

$A =$ [1]

- (c) (i) Attach the newton-meter to the string loop and lay it horizontally on the bench.

- (ii) Pull the newton-meter horizontally (parallel to the rulers). Measure and record the force F when the top ruler detaches.

$F =$ [2]

- (d) (i) Remove the Blu-Tack from the rulers.

- (ii) Make a ball with a slightly different diameter and repeat (a)(iii), (a)(iv), (b)(i), (b)(iii) and (c).

You should be aware that a large change in diameter could result in a reading outside the range of the newton-meter.

ball diameter =

$d =$ [1]

$A =$ [1]

$F =$ [2]

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- (e) It is suggested that the relationship between F and A is

$$F = kA$$

where k is a constant.

- (i) Using your data, calculate two values of k .

first value of $k = \dots\dots\dots$

second value of $k = \dots\dots\dots$

[1]

- (ii) Explain whether your results support the suggested relationship.

.....

 [1]

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- (f) (i) Describe four sources of uncertainty or limitations of the procedure for this experiment.

1.

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2.

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3.

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4.

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[4]

- (ii) Describe four improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

1.

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2.

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3.

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4.

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[4]

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